

AYE
FIT HUB

• CRF • DAY 10

Flexibility Fundamentals

Mobility for Athletes

Day 10 of 30 | Core & Flexibility

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- Flexibility vs mobility — the critical difference
- The science of how muscles stretch and adapt
- Static, dynamic, PNF & active stretching explained
- Full-body Day 10 flexibility routine (30–45 min)
- The 7 areas most people are tight — and why

01 | Flexibility vs Mobility — Two Different Things

Flexibility is the passive range of motion a muscle can achieve when an external force (gravity, a partner, or a strap) assists the movement. **Mobility** is the active range of motion a joint can achieve under its own muscular control. You can be flexible without being mobile — like a person who can touch their toes when pushed, but cannot actively lift their leg that high. Both matter. Both require training.

Quality	Definition	How to Train It	Example Test
Flexibility	Passive range of motion — how far a muscle can be lengthened	Static & PNF stretching, yoga, assisted stretches	How far you can reach in a seated forward fold with help
Mobility	Active range of motion — how far a joint moves under muscle control	Dynamic stretching, joint circles, resistance through range	How high you can actively lift your leg without assistance
Stability	Ability to control a joint at end range without collapsing	Strengthening in stretched position, balance drills	Single-leg squat depth without knee caving inward

KEY POINT — Flexibility Without Mobility Creates Injury Risk

If you stretch a muscle to a range you cannot actively control, that range becomes a "dead zone" — your nervous system cannot stabilise it under load. A hypermobile gymnast who lacks hip stability is more prone to injury than a stiffer athlete who fully controls their range. Train both flexibility AND the strength to own that range.

02 | The Science — How Your Muscles Actually Stretch

When you stretch a muscle, two key sensors respond. The **muscle spindle** detects rapid lengthening and triggers the stretch reflex — an involuntary contraction to protect the muscle from tearing. The **Golgi tendon organ (GTO)** detects sustained tension and, after 10–30 seconds, sends a signal to relax the muscle — this is called autogenic inhibition. This is why holding a stretch for at least 30 seconds works.



TECHNIQUE — Never Bounce in a Static Stretch

Bouncing (ballistic stretching) repeatedly triggers the muscle spindle, causing a protective contraction every time. The muscle never relaxes. Worse, micro-tears build up over time leading to scar tissue and reduced range. Hold still, breathe out, and let the GTO do its job. Progress comes from patience, not force.

03 | The 4 Types of Stretching — Which to Use When

STATIC

Hold still 30–60s

Post-workout cool-down

DYNAMIC

Controlled movement

Pre-workout warm-up

PNF

Contract then relax

Deep flexibility gains

ACTIVE

Muscle holds stretch

Sport-specific mobility

Four Types of Stretching — When and Why to Use Each

Type	How It Works	Duration / Reps	Best Timing	Example
Static	Hold muscle in stretched position without moving	30–60 sec hold × 2–3 sets	Post-workout or evening — when muscles are warm	Seated hamstring stretch, chest doorway stretch
Dynamic	Move through a range repeatedly with control	10–15 reps, controlled pace	Pre-workout warm-up — prepares joints for load	Leg swings, arm circles, hip circles, walking lunges
PNF	Contract muscle for 6s, relax, stretch deeper (repeat)	6s contract + 30s stretch × 3 rounds	Deep flexibility sessions, post-workout	Partner hamstring PNF, self-applied hip flexor PNF
Active Isolated	Hold stretch 2s using opposing muscle, release, repeat	10 reps × 2s holds	Pre or post-workout, sport-specific warm-up	Active straight-leg raise, active shoulder circles

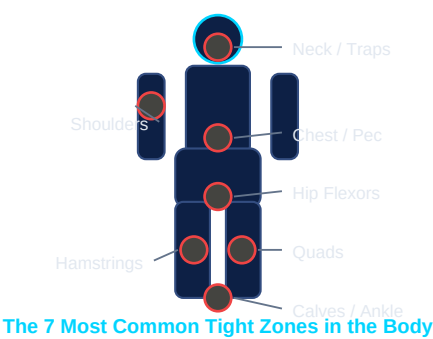
04 | Day 10 Workout — Full-Body Flexibility Routine (30–45 min)

Instructions: Hold each static stretch for 45 seconds × 2 rounds. Breathe out as you go deeper. Never stretch to pain — aim for 6–7 out of 10 tension. Do 5 deep diaphragmatic breaths before starting.

#	Stretch	Muscle Group	Hold / Reps	Technique Cue	Done ?
1	Doorway chest stretch	Pectorals, anterior shoulder	45 sec × 2	Stand in doorway, arms at 90°, lean gently forward	■
2	Overhead tricep stretch	Triceps, lats	45 sec × 2 each	Reach arm behind head, use other hand to push elbow gently	■
3	Cross-body shoulder pull	Posterior shoulder, rhomboids	45 sec × 2 each	Pull straight arm across chest, keep shoulder down	■
4	Cat-cow spinal flow	Thoracic spine, erectors	10 reps slow	Inhale on arch, exhale on round — feel each vertebra	■
5	Kneeling hip flexor lunge	Hip flexors, psoas, quads	45 sec × 2 each	Back knee down, push hips forward, keep torso upright	■
6	Pigeon pose (or figure-4)	Glutes, piriformis, IT band	60 sec × 2 each	Hinge forward from hips — do not twist spine to one side	■
7	Seated forward fold	Hamstrings, lower back	45 sec × 2	Reach for feet — flex through heels, do not round aggressively	■
8	Butterfly inner thigh	Adductors, groin	45 sec × 2	Feet together, lean forward — let gravity do the work	■
9	Standing quad stretch	Quadriceps, rectus femoris	45 sec × 2 each	Hold ankle, keep knees together, stand tall	■
10	Calf & Achilles stretch	Gastrocnemius, soleus, Achilles	30 sec each × 2	Step-back lunge on wall — bent knee hits the soleus	■

05 | The 7 Key Tight Areas — Where Most People Are Limited

Modern life — sitting at desks, looking at phones, driving — creates predictable tightness patterns in the same 7 areas for almost everyone. Knowing your tight zones lets you prioritise the right stretches rather than wasting time on areas that are already mobile.



Area	Why It Gets Tight	Priority
Neck & Traps	Screen time, forward head posture, stress tension	Side neck
Chest & Pecs	Sitting, typing, phone use shortens pectorals	Doorway thoracic
Shoulders	Repetitive overhead or forward movements	Cross-body
Hip Flexors	Sitting 6–10 hrs/day — psoas stays in shortened position	Kneeling
Hamstrings	Sitting keeps hamstrings at 90°, neuromuscular inhibition	Seated single-leg
Quads	Prolonged knee flexion, cycling, running without stretching	Kneeling femoris
Calves	Shoes with heels, running, prolonged standing	Wall calf

06 | Building Your Flexibility Practice — How to Progress

Variable	Beginner (Weeks 1–2)	Intermediate (Weeks 3–4)	Advanced (Month 2+)
Frequency	3 × per week	4–5 × per week	Daily (10–20 min minimum)
Session Length	15–20 min	25–35 min	40–60 min deep sessions
Hold Duration	20–30 sec	30–45 sec	45–60 sec + PNF rounds
Best Time	Post-workout or evening	Morning + post-workout	Dedicated flexibility sessions
What to Expect	Initial soreness, small gains in 2 weeks	Noticeable range improvement	Full-range, stable mobility across all joints

Reflection:

"Stand up right now and do a forward fold — let your arms hang. How close do your fingertips get to the floor? Now try a hip flexor lunge — can you feel the tightness? These two tests reveal your two most important flexibility targets for this 30-day program."

My flexibility targets today:

Next up — Day 11:

Meal Prep Mastery — Cook Once, Eat All Week. Build sustainable nutrition system that saves time and fuels results.

Community & support:

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